

Dataset quality assessment for speaker verification

Methods, and a structured plan to test all eight corpora

The companion log answered which datasets to use. This one answers whether they are any good — and whether they can support the feature ablation we actually want to run.

2026-08-10

methodology + plan, nothing executed yet

companion: 2026-08-10-open-si-ta-sv-corpora.md

I What quality means here

The instinct is to reach for audio-quality metrics. For speaker verification that is the least important axis. A corpus of clean studio speech can be worthless for verification, and a corpus of noisy YouTube audio can be excellent. Four things actually decide whether an SV dataset is sound — roughly in order of how badly they bite.

#	FAILURE MODE	WHAT IT LOOKS LIKE	WHY IT IS FATAL
1	Label noise	Two people sharing one speaker id, or one person split across two ids	Corrupts targets <i>and</i> impostors. A mislabelled impostor that is really a target puts a hard ceiling on measurable EER. Removing noisy VoxCeleb2 samples alone bought ~5.9 % relative improvement in one study.
2	Session / channel confound	Target pairs cut from the same recording	The model scores channel, not speaker. 8-17 % of VoxCeleb1-H target pairs have this. EER looks great and means nothing.
3	Insufficient power	20-50 speakers, however many trials	Confidence intervals wider than the effects being ranked. This is the one that decides whether our ablation is worth running at all.
4	Distribution shift / bias	Trained on Indian Tamil, deployed on Sri Lankan Tamil	Headline EER hides subgroup failure. Female error rates 49 % higher than male in one challenge-wide audit.

Audio quality matters mainly as a confound detector. If corpus A is uniformly cleaner than corpus B, an embedding separates them trivially, and any cross-corpus generalisation result is measuring codec, not voice.

The assessment is therefore six layers, cheapest and most decisive first. **Layers 0-2 need no GPU and no model.** Layer 3 is where the real answers are.

II The methods, layer by layer

Layer 0 — Integrity and inventory

Is the data physically what the metadata claims?

CHECK	METHOD	RED FLAG
Decodability	soundfile.read every file	any failure
Format uniformity	sample rate, channels, bit depth histogram	more than one value per corpus
Near-silent files	RMS below -60 dBFS over the whole file	> 0.1 % of files
Exact duplicates	MD5 over <i>decoded</i> PCM, not the container	any cross-speaker duplicate
Effective bandwidth	mean power spectrum; rolloff where energy falls 50 dB below peak	rolloff far below Nyquist → upsampled narrowband audio

The bandwidth check is the one people skip and regret. Audio resampled up from 8 kHz telephone still carries a 16000 header; the model sees a brick wall at 4 kHz and learns *this is corpus X*. slr65_tamil was resampled 48→16 kHz here, so its ceiling is 8 kHz by construction — expected and fine. We are hunting *unexpected* band-limiting and internal inconsistency.

Layer 1 — Signal quality, non-intrusively

How clean is the audio, and is cleanliness confounded with speaker identity?

No clean reference exists for any of these corpora, so every metric must be reference-free: WADA-SNR or NIST STNR; clipping rate; speech ratio via VAD; LUFS loudness; and **torchaudio.pipelines.SQUIM_OBJECTIVE / SQUIM_SUBJECTIVE**, which estimate STOI, PESQ, SI-SDR and MOS with no reference. SQUIM is chosen over DNSMOS and NISQA purely because it ships inside the already-installed torchaudio 2.8 and adds no dependency.

The critical analysis is not the mean — it is the variance within a speaker. If a speaker's utterances all share one SNR and one loudness while different speakers differ, then SNR is speaker id and the model can cheat. Compute the between- vs within-speaker variance ratio (a one-way ANOVA F) for each signal metric. A high F on SNR warns that target trials are partly solvable from channel alone.

Layer 2 — Distribution and coverage

Is the speaker population shaped so a meaningful trial list can even be drawn?

- Speakers; utterances per speaker (min / median / max); **Gini coefficient** of utterances per speaker — imbalance in one number.
- **Duration histogram, and the fraction below 2 s and 4 s.** SV accuracy collapses on short utterances; a corpus with a 2.5 s median reports a bad EER for reasons unrelated to its speakers.
- Sessions per speaker, and how many speakers have ≥ 2 — this decides how many can supply cross-session targets at all.
- Gender balance, and gender $\times$ utterance count.
- Duration $\times$ speaker correlation — a confound if a few speakers own all the long files.

All of this is computable from metadata/utterances.csv, which every dataset folder already carries.

Layer 2 costs minutes and needs no audio.

Layer 3 — Label reliability via speaker embeddings

Are the speaker labels actually correct? This is the layer that finds real problems.

- **Extract one embedding per utterance** with a strong pretrained model, ideally two architecturally different ones so findings are not model artefacts. tools/zeroshot_eval.py already does extraction with caching.
- **Within- vs between-speaker similarity.** Plot both cosine distributions. Heavy overlap means either hard data or broken labels — and layers 4–5 cannot tell you which.
- **Leave-one-out centroid outliers.** For each utterance, cosine to its own speaker's centroid computed without it; rank ascending. The bottom tail is the mislabel candidate list. **Then actually listen to the top ~30** — this step is not automatable and is the only way to turn a suspicion into a fact.

- **Per-speaker silhouette score.** Low silhouette means acoustically unremarkable or two people in one id.
- **Clustering vs labels.** Agglomerative clustering (cosine, average linkage), compared to labels via NMI, ARI, V-measure, purity. Clusters >> speakers → one person split across ids. Clusters << speakers → distinct ids collapsing, i.e. probable duplicates.
- **Visual check** with t-SNE or UMAP on a speaker subsample.

Thresholds must be set per corpus type, not globally. Read speech recorded in one sitting should cluster almost perfectly (NMI > 0.95); anything less is a real label problem. In-the-wild corpora legitimately score lower, so the same number means different things.

Layer 4 — Protocol and trial-list audit

Is the benchmark we built out of this corpus honest? A trial list can be wrong in ways no model will ever reveal.

CHECK	REQUIREMENT
Target / impostor counts and ratio	matches the design
Self-pairs (a == b)	zero
Duplicate unordered pairs	zero
Same-(speaker, session) target pairs	zero, wherever sessions are real
Same-gender impostor fraction	1.0 where gender is known; documented otherwise
Speaker overlap between train_list and test_list	zero, or explicitly declared
Utterance reuse frequency	no small set of files dominating trials
Enrol / test duration matching	no systematic asymmetry

The train/test overlap check becomes essential the moment we *combine* corpora — training on SLR127 + Kathbath and evaluating on a merged list is exactly where leakage sneaks in. Note a subtlety already hit in this repo: comparing session *directory names* is wrong when session names are shared across speakers (SLR127 uses ISTL/MICI/MILE). The key must be the (speaker, session) pair.

Layer 5 — Difficulty calibration and statistical power

How hard is this benchmark, and can it detect the effects we care about?

Difficulty. Run 2–3 pretrained models zero-shot; report EER_avg, EER_max and minDCF at p_target 0.01 and 0.05 — all of which tools/zeroshot_eval.py already computes. Express results as a ratio against the same models' VoxCeleb-O numbers for a corpus-independent difficulty index. Report **Cllr** and **min Cllr** too; the gap between them is miscalibration, which our earlier finding on cross-language calibrator swaps (Cllr inflated up to 5.7×) makes directly relevant.

Power is the decisive analysis, and the bootstrap must be speaker-clustered. Resample *speakers* with replacement — not trials — rebuild the trial subset, recompute EER, repeat B = 1000, take the 2.5 / 97.5 percentiles. Resampling trials independently is wrong because trials from one speaker are correlated, and it understates the interval, often badly when speakers are few.

From that, derive the **minimum detectable effect (MDE)**: the smallest EER difference the corpus can resolve at 80 % power, $\alpha = 0.05$, for a paired comparison on identical trials. Then state plainly whether the ablation deltas we expect (~0.2–0.5 % EER) exceed it. This turns *50 speakers feels too few* into a number, and it is the single most useful output of the exercise.

Layer 6 — Bias and cross-corpus transfer

Whose voices does this work for, and does it survive a domain change?

- **Subgroup EER** by gender and, where known, dialect — reporting per-subgroup FMR/FNMR at a *shared* threshold, not just per-subgroup EER, because a single global operating point is what a deployed system actually uses.

- **Fairness Discrepancy Rate (FDR)** for a one-number summary, following Hutiri & Ding's bias-quantification framework.
- **Cross-corpus matrix:** train on A, evaluate on B for every ordered pair. The Indian-Tamil-train / Sri-Lankan-Tamil-test cell is the number this project is positioned to publish.
- **Channel probe:** train a logistic regression on embeddings to predict corpus-of-origin. Near-perfect accuracy means the embedding encodes channel as much as identity.

Reporting standard

Findings belong in each corpus's existing metadata.json and README.md rather than a separate report — the datasheet lives with the data. Framing follows **Datasheets for Datasets** (Gebru et al.) and **Data Statements** (Bender & Friedman). Our metadata.json already covers provenance, licence, session semantics and caveats; layers 0-6 add the *measured* half.

III Structured test plan

Phasing

Ordered so the cheapest checks can kill a corpus before expensive ones run.

PHASE	LAYERS	COMPUTE	WALL-CLOCK	GATE
P0	L0 + L2	CPU, metadata only	~2 h	integrity clean, distribution documented
P1	L4	CPU, lists only	~2 h	no leakage, no same-session targets
P2	L1	CPU (GPU faster)	~1 day	quality/speaker confound quantified
P3	L3	GPU	~2 days	label noise bounded, listened-to sample
P4	L5	GPU (reuses P3)	~1 day	MDE per corpus → go/no-go for the ablation
P5	L6	GPU	~2 days	subgroup + cross-corpus gaps reported

P0 and P1 need nothing beyond this node. P3 onward needs the GPU offload path (tools/gpurun.sh — this node has no GPU); NFS is mounted at the same path on the compute nodes, so no data moves. Embeddings extracted in P3 are cached and reused by P4 and P5 — extract once.

Per-dataset plan

Every corpus gets L0/L2/L4. The rest is targeted at each corpus's specific known risk, so effort goes where it can change a decision.

slr52_sinhala

HIGH

478 spk · 185k utts · 224.5 h

Risk. The entire Sinhala side rests on it. Speaker ids are anonymised crowdsourcing accounts with **no upstream guarantee that one account is one person**; no gender labels; single sitting.

Run. Full L0; L2 with attention to the duration tail; **L3 in full** — this is where a clustering-vs-labels check is most valuable, precisely because nothing guarantees account = person. Expect NMI > 0.95. L1 on a 5k-utterance sample. **Red flag that would change plans:** clusters >> speakers, implying shared or reused accounts.

slr127_tamil

HIGH

531 spk · 89k utts · 150.1 h

Risk. Largest Tamil pool and the only read-speech corpus with real sessions. But the speaker-id convention was **derived by us, not documented upstream**, and prefix-as-session is an inference.

Run. Full L0 including the **bandwidth check** (three batches may differ in equipment); **L3 with clustering split by prefix** — this directly tests the session hypothesis: if ISTL and MILE recordings of one speaker cluster apart, the prefix is a genuine channel change, validating the mapping *and* quantifying the effect. L1 **stratified by prefix** is the physical evidence. This corpus deserves the most careful L3 because our confidence rests on inference, not documentation.

kathbath_tamil

MED-HIGH

60 spk · 7k utts · 13.2 h

Risk. The comparability anchor. Lossy AAC; only 20 speakers per eval split; 11 numeric ids recur across splits, namespaced apart on an **unverified assumption** that they are different people.

Run. Full L0/L2/L4; **L3 with one extra test — embed kbv84, kbk84, kbt84 and check whether same-numeric-id speakers across splits are in fact the same person.** High cosine similarity would mean our namespacing silently created three identities for one speaker. Concrete, falsifiable, worth answering early. L5 matters here: 20 speakers with 50k trials is the textbook case of trial count overstating power. **Do not modify the official lists** on any finding — document.

nisp_tamil**MEDIUM**

65 spk (all bilingual) · 4.9k utts · 13.3 h

Risk. Small, but the only cross-lingual probe. English is accented, not a native control.**Run.** L0/L2/L4; **L3 focused on the cross-lingual question** — compare within-speaker-within-language, within-speaker-cross-language, and between-speaker similarity distributions. The gap between the first two is the language effect on the embedding, measured directly, with no trial list needed. L6 is unusually feasible: age, height and region ship with the corpus.**slr65_tamil****LOW-MED**

50 spk · 4.3k utts · 7.1 h

Risk. Kept for v0 continuity only.**Run.** L0/L2/L4 for completeness; L3 cheap because it is small; **L5 to put a number on how underpowered v0 was.** That number retroactively frames every v0 result and belongs in the thesis. Not worth deep L1/L6 investment.**commonvoice_tamil****MEDIUM**

not yet fetched

Risk. client_id is an *account*, not verifiably a person — the highest label-noise risk in the collection. Wildly heterogeneous devices.**Run.** L0 with **bandwidth check as a first-class concern**; L1 expecting high variance — and here high variance is a *feature*, since device diversity is why we want it; **L3 mandatory before any use**, with the account-vs-person question explicit. L2 should set --min-utts empirically rather than the current default of 4.**sita_sinhala_tamil****MEDIUM**

not yet fetched

Risk. For diarization and domain probing; **not** an SV benchmark.**Run.** L0/L1/L2 only. **L4 and L5 do not apply** — there is no trial list, by design. One bespoke analysis is worth doing: **cross-recording identity linking**. Embed all segments, cluster across recordings, and see whether the same public figures recur. If they do, that is the raw material for the first genuine Sri Lankan SV benchmark — and the measurement tells you whether the manual effort is worth it. That is a research contribution, not a QA step.**slceleb_sinhala_tamil****HIGHEST**

audio not on mounts

Risk. The reference set, and the only corpus whose absolute EER means what a reader will assume. YouTube-derived pipelines are exactly where label noise lives — VoxCeleb's lineage proves it — and unlike VoxCeleb1 this corpus has **no published cleaning pass**.**Run.** **All six layers, in full.** Nothing else carries as much weight per speaker. L3 here is not optional. The L6 cross-corpus matrix against SLR127/Kathbath is the headline experiment.**Acceptance gates***To be revised once P0/P2 show what the corpora actually look like — and split by corpus type, because read speech and in-the-wild are not comparable.*

GATE	READ-SPEECH	IN-THE-WILD
Decode failures	0	0
Cross-speaker exact duplicates	0	0
Utterances < 2 s	< 5 %	< 10 %
Label NMI (AHC vs labels)	> 0.95	> 0.80

Centroid-outlier tail flagged for listening	bottom 0.5 %	bottom 1 %
Same-(speaker, session) target pairs	0	0
Train/test speaker overlap	0	0
Same-gender impostor fraction	1.0 where known	1.0 where known

A corpus failing a gate is **not discarded** — it is documented and down-weighted, and the failure goes into its metadata.json caveats.

Deliverables

- **tools/dataset_qc.py** — L0/L2/L4, pure CPU, emits metadata/qc_report.json per dataset plus a cross-corpus summary.
- **tools/signal_quality.py** — L1: WADA-SNR, clipping, VAD, SQUIM, sampled.
- **tools/label_audit.py** — L3: consumes cached embeddings from zeroshot_eval.py; emits outlier lists, clustering metrics, and a listen-to-me shortlist of paths.
- **tools/eer_power.py** — L5: speaker-clustered bootstrap CIs and MDE.
- A results log plus a quality block added to each corpus's metadata.json.

What would change the ablation plan

The point of all this is one decision. Three outcomes are actionable:

IF MDE IS...	THEN
below ~0.2 % EER	the ablation is properly powered and proceeds as designed
0.3 – 0.5 % EER	only large feature effects are rankable — narrow the ablation to fewer, bolder conditions rather than a wide grid
above 0.5 % EER	no amount of clever features produces a defensible ranking; the honest move is more speakers before more experiments

Layer 3 can also invalidate a corpus outright. Better to find that in P3 than in peer review.

IV Environment notes

Verified on this host, 2026-08-10.

- torch 2.8.0+cu128, torchaudio 2.8.0 — **SQUIM_OBJECTIVE** and **SQUIM_SUBJECTIVE** import cleanly.
- sklearn 1.5.1 — TSNE, AgglomerativeClustering, silhouette, NMI/ARI all present.
- speechbrain present; wespeaker, pyannote.audio and umap-learn absent (all optional).
- **No GPU on this node** (torch.cuda.is_available() is False). L3-L6 go through tools/gpurun.sh; NFS is mounted at the same path on compute nodes, so no data moves.
- Reusable already: tools/zeroshot_eval.py (cached embedding extraction, EER/minDCF), tools/check_wav_integrity.py, tools/sl_dataprep.py.

V References

Hutiri, Ding (2022)	Bias in Automated Speaker Recognition. FAccT 2022. arXiv:2201.09486
Hutiri et al. (2022)	Interspeech — same-recording confound in VoxCeleb trial pairs
Kumar et al. (2023)	TorchAudio-Squim: Reference-less Speech Quality and Intelligibility. arXiv:2304.01448
Gebbru et al. (2021)	Datasheets for Datasets. CACM 64(12)
Bender, Friedman (2018)	Data Statements for NLP. TACL 6
Tong et al. (2022)	Inconsistency Ranking-based Noisy Label Detection. arXiv:2212.00239

CEC (2024)	A Noisy Label Detection Method for Speaker Recognition. arXiv:2406.13268
Fathan et al. (2025)	Automatic Labeling and Correction of Noisy Labels for Robust SSL Speaker Verification. Interspeech 2025
Bisani, Ney (2004)	Bootstrap Estimates for Confidence Intervals in ASR Performance Evaluation. ICASSP 2004
Chung et al. (2019)	VoxSRC 2019. arXiv:1912.02522 — dev/test overlap across VoxCeleb1/2/SITW
NIST SRE	evaluation plans — reference for trial protocol and DCF reporting
BOSARIS toolkit	Cllr / min Cllr calibration metrics